Girsanov and Risk-Neutral Pricing

Lecture 6 Notes — Densities, Drift Changes, and Risk-Neutral Dynamics

Route through Lecture 6

A rare Gaussian tail first motivates reweighting. Finite scenarios and Gaussian tilting explain the density; exponential weights extend the idea to Brownian paths. Girsanov identifies the new Brownian motion, and the Black–Scholes example chooses its shift so that the discounted stock has no drift. Lecture 7 starts from this result and develops replication, conditional-expectation pricing, and completeness.

1 Densities and Bayes' Rule

1.1 A rare Gaussian tail motivates reweighting

Let $X \sim N(0,1)$ under $\mathbb{P}$, with Φ the standard normal CDF, and consider

$$p = \mathbb{P}(X > 5) = 1 - \Phi(5) \approx 2.87 \times 10^{-7}.$$

The direct Monte Carlo estimate from n independent draws is $\widehat{p}_{\mathrm{MC}} = n^{-1} \sum_{i=1}^{n} \mathbf{1}_{\{X_i > 5\}}$. Even for $n = 10^6$, the expected number of tail observations is only $np \approx 0.29$. The probability of seeing none is $(1-p)^n \approx 0.75$, in which case the estimate is zero.

To visit the tail more often, sample under a new law $\mathbb{Q}$ for which the same variable has distribution $X \sim N(5,1)$. Then $\mathbb{Q}(X > 5) = 1/2$. This $\mathbb{Q}$ is chosen for simulation; it need not be a pricing measure. The likelihood correction that recovers p is derived in the Gaussian tilting section below.

1.2 Finite spaces reveal the mechanism

Let $\Omega = \{\omega_1, \dots, \omega_m\}$, and suppose $\mathbb{P}(\omega_i) > 0$ and $\mathbb{Q}(\omega_i) > 0$. Define the density ratio

$$Z(\omega_i) = \frac{\mathbb{Q}(\omega_i)}{\mathbb{P}(\omega_i)}.$$

Then

$$Z > 0, \qquad \mathbb{E}^{\mathbb{P}}[Z] = 1, \qquad \boxed{\mathbb{E}^{\mathbb{Q}}[Y] = \mathbb{E}^{\mathbb{P}}[ZY].}$$

A state with $Z > 1$ is upweighted and one with $Z < 1$ is downweighted.

Example — a three-state reweighting

Consider

	ω_1	ω_2	ω_3
$\mathbb{P}$	$1/2$	$1/3$	$1/6$
$\mathbb{Q}$	$1/4$	$1/2$	$1/4$

Then

$$Z(\omega_1) = \frac{1}{2}, \qquad Z(\omega_2) = Z(\omega_3) = \frac{3}{2},$$

and indeed

$$\mathbb{E}^{\mathbb{P}}[Z] = \frac{1}{2}\frac{1}{2} + \frac{1}{3}\frac{3}{2} + \frac{1}{6}\frac{3}{2} = 1.$$

For $Y = (-2, 1, 4)$, direct computation and reweighting agree:

$$\mathbb{E}^{\mathbb{Q}}[Y] = 1 = \mathbb{E}^{\mathbb{P}}[ZY].$$

1.3 Radon–Nikodym densities and equivalence

Definition 1.1 (Change of measure by a density)**.** Let $Z \geq 0$ be $\mathcal{F}_T$-measurable with $\mathbb{E}^{\mathbb{P}}[Z] = 1$. Define

$$\mathbb{Q}(A) = \mathbb{E}^{\mathbb{P}}[Z\mathbf{1}_A], \qquad A \in \mathcal{F}_T.$$

Then $\mathbb{Q}$ is a probability measure and

$$Z = \frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}\bigg|_{\mathcal{F}_T}.$$

For every $\mathbb{Q}$-integrable Y,

$$\boxed{\mathbb{E}^{\mathbb{Q}}[Y] = \mathbb{E}^{\mathbb{P}}[ZY].}$$

Absolute continuity $\mathbb{Q} \ll \mathbb{P}$ means that a $\mathbb{P}$-null event remains impossible under $\mathbb{Q}$. If $Z > 0$ $\mathbb{P}$-almost surely, then $\mathbb{P} \sim \mathbb{Q}$: both measures have exactly the same null events. In that case

$$\frac{\mathrm{d}\mathbb{P}}{\mathrm{d}\mathbb{Q}} = \frac{1}{Z} \quad \mathbb{Q}\text{-a.s.}, \qquad \mathbb{E}^{\mathbb{Q}}\left[\frac{1}{Z}\right] = 1.$$

Three properties with different roles

- $Z \geq 0$ makes $\mathbb{Q}(A) = \mathbb{E}^{\mathbb{P}}[Z\mathbf{1}_A]$ nonnegative.
- $\mathbb{E}^{\mathbb{P}}[Z] = 1$ gives total mass one.
- $Z > 0$ gives equivalence, which prevents the change of measure from deleting possible scenarios.

Do not replace one property by another.

1.4 Conditional normalization and the density process

Let $\mathcal{G} \subseteq \mathcal{F}_T$. If the denominator is positive, Bayes' rule under a change of measure is

$$\boxed{\mathbb{E}^{\mathbb{Q}}[Y \mid \mathcal{G}] = \frac{\mathbb{E}^{\mathbb{P}}[ZY \mid \mathcal{G}]}{\mathbb{E}^{\mathbb{P}}[Z \mid \mathcal{G}]}.}$$

The numerator reweights future outcomes. The denominator renormalizes those weights after the information in $\mathcal{G}$ is known.

To prove the formula, set

$$X = \frac{\mathbb{E}^{\mathbb{P}}[ZY \mid \mathcal{G}]}{\mathbb{E}^{\mathbb{P}}[Z \mid \mathcal{G}]}.$$

The variable X is $\mathcal{G}$-measurable. For every $A \in \mathcal{G}$,

$$\begin{aligned}
\mathbb{E}^{\mathbb{Q}}[\mathbf{1}_A X] &= \mathbb{E}^{\mathbb{P}}[Z\mathbf{1}_A X] \\
&= \mathbb{E}^{\mathbb{P}}[\mathbf{1}_A X \mathbb{E}^{\mathbb{P}}[Z \mid \mathcal{G}]] \\
&= \mathbb{E}^{\mathbb{P}}[\mathbf{1}_A \mathbb{E}^{\mathbb{P}}[ZY \mid \mathcal{G}]] \\
&= \mathbb{E}^{\mathbb{P}}[\mathbf{1}_A ZY] = \mathbb{E}^{\mathbb{Q}}[\mathbf{1}_A Y].
\end{aligned}$$

This is the defining test for $X = \mathbb{E}^{\mathbb{Q}}[Y \mid \mathcal{G}]$.

Given a terminal density Z_T, define

$$\boxed{Z_t = \mathbb{E}^{\mathbb{P}}[Z_T \mid \mathcal{F}_t].}$$

Then $Z = (Z_t)_{0 \leq t \leq T}$ is the density process, and

$$\boxed{\mathbb{E}^{\mathbb{Q}}[Y \mid \mathcal{F}_t] = \frac{1}{Z_t}\mathbb{E}^{\mathbb{P}}[Z_T Y \mid \mathcal{F}_t].}$$

Information is not changed by reweighting

If Y is already $\mathcal{G}$-measurable, then

$$\mathbb{E}^{\mathbb{P}}[ZY \mid \mathcal{G}] = Y\mathbb{E}^{\mathbb{P}}[Z \mid \mathcal{G}],$$

so $\mathbb{E}^{\mathbb{Q}}[Y \mid \mathcal{G}] = Y$. Changing measure changes probabilities, not what has already been observed.

2 Gaussian Tilting: the Finite-Dimensional Prototype

2.1 Completing the square fixes the sign

Let $X \sim N(0,1)$ under $\mathbb{P}$, and define

$$Z = \exp\left(-\theta X - \frac{1}{2}\theta^2\right).$$

The Gaussian moment-generating function gives

$$\mathbb{E}^{\mathbb{P}}[Z] = e^{-\theta^2/2}\mathbb{E}^{\mathbb{P}}[e^{-\theta X}] = 1.$$

Under $\mathbb{Q}$, the density of X is

$$e^{-\theta x - \theta^2/2}\frac{e^{-x^2/2}}{\sqrt{2\pi}} = \frac{1}{\sqrt{2\pi}}e^{-(x+\theta)^2/2}.$$

Therefore

$$\boxed{X \sim N(-\theta, 1) \text{ under } \mathbb{Q}.}$$

For $\theta > 0$, the factor $e^{-\theta X}$ favors negative values. This quick qualitative check is often enough to detect a sign error. More generally, if $X \sim N(0, \sigma^2)$ and we want mean m under $\mathbb{Q}$, take

$$\boxed{Z = \exp\left(\frac{m}{\sigma^2}X - \frac{m^2}{2\sigma^2}\right).}$$

Then $X \sim N(m, \sigma^2)$ under $\mathbb{Q}$. The mean changes; the variance does not.

Exercise — reverse-engineer the normalizer

Suppose $X \sim N(0, v)$ and $Z = e^{aX+b}$. Find the unique b for which $\mathbb{E}[Z] = 1$, then identify the law of X under $\mathrm{d}\mathbb{Q} = Z\,\mathrm{d}\mathbb{P}$.

Short solution

Since $\mathbb{E}[e^{aX}] = e^{a^2v/2}$, normalization requires $b = -a^2v/2$. Completing the square gives

$$\boxed{X \sim N(av, v) \text{ under } \mathbb{Q}.}$$

The linear coefficient controls the new mean; the quadratic compensator ensures total mass one.

2.2 Reweighting and translation

For every bounded measurable f,

$$\boxed{\mathbb{E}^{\mathbb{Q}}[f(X)] = \mathbb{E}^{\mathbb{P}}[Zf(X)] = \mathbb{E}^{\mathbb{P}}[f(X-\theta)]}$$

in the standard-normal example above. The first equality is the definition of the new measure. The second is a Gaussian distributional identity. It is important not to confuse these two logical steps.

What survives in continuous time

A Brownian path is an infinite collection of compatible Gaussian variables. The continuous-time density will again contain

$$\text{linear Gaussian exposure} - \frac{1}{2}\text{variance of that exposure.}$$

This is the origin of the stochastic exponential.

2.3 A shifted tail with and without likelihood weights

Return to $p = \mathbb{P}(X > 5)$ with $X \sim N(0,1)$ under $\mathbb{P}$. The Gaussian tilt with target mean 5 gives $X \sim N(5,1)$ under $\mathbb{Q}$, with

$$\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}(x) = e^{5x-25/2}, \qquad w(x) = \frac{\mathrm{d}\mathbb{P}}{\mathrm{d}\mathbb{Q}}(x) = e^{-5x+25/2}.$$

For independent draws $X_1, \ldots, X_n$ under $\mathbb{Q}$, merely shifting the sampling mean gives

$$\frac{1}{n}\sum_{i=1}^{n} \mathbf{1}_{\{X_i>5\}} \longrightarrow \mathbb{Q}(X > 5) = \frac{1}{2}.$$

This is the tail probability under the new law. To estimate the original probability, reweight each observation:

$$\boxed{\widehat{p}_{\mathrm{IS}} = \frac{1}{n}\sum_{i=1}^{n} w(X_i)\mathbf{1}_{\{X_i>5\}}, \qquad \mathbb{E}^{\mathbb{Q}}[\widehat{p}_{\mathrm{IS}}] = \mathbb{E}^{\mathbb{Q}}[w(X)\mathbf{1}_{\{X>5\}}] = \mathbb{P}(X > 5) = p.}$$

This is importance sampling: the shift makes tail observations frequent, and the inverse density restores their original probability weight.

3 Why Change the Probability Measure?

3.1 One random payoff, two systems of weights

Let $H : \Omega \to \mathbb{R}$ be a terminal payoff. A probability measure assigns weights to the scenarios $\omega \in \Omega$; it does not alter the value $H(\omega)$. If $\mathbb{P}$ is replaced by an equivalent measure $\mathbb{Q}$, then the sample space, filtration, stock paths, and payoff remain the same. Only their probabilities change.

Physical and pricing viewpoints

- Under $\mathbb{P}$, parameters are interpreted statistically: they describe frequencies or forecasts in the model.
- Under $\mathbb{Q}$, probabilities are selected so that prices expressed in units of a chosen numeraire are martingales.
- A pricing measure is therefore constrained by traded prices. It is not a claim about investor beliefs or the most likely real-world outcome.

Suppose, for the moment, that the stock and bank account satisfy under $\mathbb{P}$

$$\frac{\mathrm{d}S_t}{S_t} = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}W_t, \qquad \mathrm{d}B_t = rB_t\,\mathrm{d}t, \qquad B_t = e^{rt}.$$

The discounted stock is $\bar{S}_t = S_t/B_t = e^{-rt}S_t$. Since the discount factor has finite variation, the product rule gives

$$\boxed{\frac{\mathrm{d}\bar{S}_t}{\bar{S}_t} = (\mu - r)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t.}$$

Thus the term to remove is the excess drift $\mu - r$. Measured per unit of volatility, it is

$$\lambda = \frac{\mu - r}{\sigma},$$

the market price of Brownian risk in this one-factor model.

Exercise — discount before changing measure

Let $B_t = \exp(\int_0^t r_s\,\mathrm{d}s)$ and $\mathrm{d}S_t/S_t = \mu_t\,\mathrm{d}t + \sigma_t\,\mathrm{d}W_t$. Derive the SDE of S_t/B_t without assuming constant coefficients.

Short solution

Writing $D_t = B_t^{-1}$, we have $\mathrm{d}D_t = -r_t D_t\,\mathrm{d}t$ and $[D, S] = 0$. Hence

$$\boxed{\frac{\mathrm{d}(D_t S_t)}{D_t S_t} = (\mu_t - r_t)\,\mathrm{d}t + \sigma_t\,\mathrm{d}W_t.}$$

Discounting identifies exactly which drift Girsanov must eliminate.

Expectation under $\mathbb{P}$ is not an arbitrage-free price

In general,

$$e^{-r(T-t)}\mathbb{E}^{\mathbb{P}}[H \mid \mathcal{F}_t]$$

is neither a replication price nor a no-arbitrage price. The physical drift and risk preferences can enter that expectation. Pricing by expectation only becomes valid after specifying the appropriate pricing measure and numeraire.

4 From Gaussian Tilting to Brownian Paths

4.1 A deterministic path exposure

Let W be Brownian motion under $\mathbb{P}$, and let $h \in L^2([0,T])$ be deterministic. The stochastic integral

$$X = \int_0^T h_s\,\mathrm{d}W_s$$

is centered Gaussian with variance

$$v = \int_0^T h_s^2\,\mathrm{d}s.$$

Consequently,

$$\boxed{Z_T = \exp\left(\int_0^T h_s\,\mathrm{d}W_s - \frac{1}{2}\int_0^T h_s^2\,\mathrm{d}s\right)}$$

satisfies $\mathbb{E}^{\mathbb{P}}[Z_T] = 1$.

Theorem 4.1 (Cameron–Martin drift shift)**.** Define $\mathbb{Q}$ on $\mathcal{F}_T$ by $\mathrm{d}\mathbb{Q} = Z_T\,\mathrm{d}\mathbb{P}$. Then

$$\boxed{B_t = W_t - \int_0^t h_s\,\mathrm{d}s}$$

is Brownian motion under $\mathbb{Q}$. Equivalently,

$$\boxed{\mathrm{d}W_t = \mathrm{d}B_t + h_t\,\mathrm{d}t.}$$

The density rewards paths whose increments align with h. Under the new weights, W has accumulated drift $\int_0^t h_s\,\mathrm{d}s$, and subtracting that drift recovers a Brownian motion.

Example — constant exposure

If $h_t \equiv \theta$, then

$$Z_T = e^{\theta W_T - \theta^2 T/2}, \qquad B_t = W_t - \theta t.$$

Thus $W_T = B_T + \theta T$, and

$$\boxed{W_T \sim N(\theta T, T) \text{ under } \mathbb{Q}.}$$

This is precisely Gaussian tilting with a variance T.

4.2 Why an adapted exposure is harder

If h_t is replaced by a random progressively measurable process ϕ_t, then $\int_0^T \phi_s \, dW_s$ need not be Gaussian. Conditional on the entire path of ϕ, the integral is not generally an independent normal random variable, because ϕ may depend on the same Brownian motion. The single Gaussian-mgf argument therefore breaks down.

This is not a problem of notation. It is the reason we need stochastic calculus and a genuine martingale condition.

5 The Stochastic Exponential and Its Validity

5.1 The compensation term is forced by Itô

Let ϕ be progressively measurable and suppose

$$\int_0^T \phi_s^2 \, ds < \infty \quad \text{almost surely.}$$

Define

$$M_t = \int_0^t \phi_s \, dW_s, \qquad [M]_t = \int_0^t \phi_s^2 \, ds,$$

and its stochastic exponential

$$\boxed{Z_t = \mathcal{E}(M)_t = \exp\left(M_t - \frac{1}{2}[M]_t\right).}$$

Set $Y_t = M_t - \frac{1}{2}[M]_t$, so $Z_t = e^{Y_t}$. Since

$$dY_t = \phi_t \, dW_t - \frac{1}{2}\phi_t^2 \, dt, \qquad d[Y]_t = \phi_t^2 \, dt,$$

Itô's formula gives

$$\begin{aligned} dZ_t &= Z_t \, dY_t + \frac{1}{2} Z_t \, d[Y]_t \\ &= Z_t \phi_t \, dW_t - \frac{1}{2} Z_t \phi_t^2 \, dt + \frac{1}{2} Z_t \phi_t^2 \, dt. \end{aligned}$$

Therefore

$$\boxed{dZ_t = Z_t \phi_t \, dW_t, \qquad Z_0 = 1.}$$

The coefficient $(1/2)$ is exactly what cancels the quadratic-variation drift.

Exercise — detect the only valid compensator

For $c \in \mathbb{R}$, define

$$Z_t^{(c)} = \exp\left(\int_0^t \phi_s \, dW_s - c \int_0^t \phi_s^2 \, ds\right).$$

Find the drift of $Z^{(c)}$ and determine the unique c that makes it a local martingale for every admissible ϕ.

Short solution

Itô's formula gives

$$\frac{dZ_t^{(c)}}{Z_t^{(c)}} = \phi_t \, dW_t + \left(\frac{1}{2} - c\right) \phi_t^2 \, dt.$$

Hence

$$\boxed{c = \frac{1}{2}.}$$

The correction is not a convention; it is forced by quadratic variation.

5.2 Local martingale is not enough

The stochastic exponential is positive and is a local martingale. Every positive local martingale is a supermartingale, so

$$\mathbb{E}^{\mathbb{P}}[Z_t] \leq Z_0 = 1.$$

It can lose mass. If $\mathbb{E}^{\mathbb{P}}[Z_T] < 1$, the terminal variable cannot define a probability measure of total mass one.

The most common unjustified step

Writing $\mathrm{d}Z_t = Z_t \phi_t \, \mathrm{d}W_t$ proves only that Z is a local martingale. It does not prove $\mathbb{E}[Z_T] = 1$. Before defining $\mathbb{Q}$, one must verify a true-martingale condition or explicitly assume it.

Theorem 5.1 (Novikov's sufficient condition)**.** If

$$\boxed{\mathbb{E}^{\mathbb{P}}\left[\exp\left(\frac{1}{2}\int_0^T \phi_s^2 \, \mathrm{d}s\right)\right] < \infty,}$$

then $Z = \mathcal{E}(\int \phi \, \mathrm{d}W)$ is a true martingale on ([0,T]), and $\mathbb{E}^{\mathbb{P}}[Z_T] = 1$.

If $|\phi_t| \leq C$, then

$$\int_0^T \phi_s^2 \, \mathrm{d}s \leq C^2 T,$$

and Novikov follows immediately. By contrast, almost-sure finiteness of $\int \phi^2$ does not automatically control its exponential moment.

Example — a bounded adapted exposure

Fix $t_0 < T$ and let ξ be bounded and $\mathcal{F}_{t_0}$-measurable. Set

$$\phi_t = \xi \mathbf{1}_{(t_0, T]}(t).$$

This process is progressively measurable, and

$$\int_0^T \phi_t^2 \, \mathrm{d}t = \xi^2 (T - t_0) \leq \|\xi\|_\infty^2 T.$$

Novikov holds even though the drift shift is random. Economically, the future change of drift may depend on information already observed at time t_0, but not on future increments.

Localization separates algebra from validity

If ϕ is unbounded, define stopping times

$$\tau_n = \inf\left\{t : \int_0^t \phi_s^2 \, \mathrm{d}s \geq n\right\} \wedge T.$$

The stopped integrand $\phi_t \mathbf{1}_{\{t \leq \tau_n\}}$ has bounded accumulated energy, so its stochastic exponential is a true martingale. This validates the drift calculation up to τ_n. Passing to the unstopped limit still requires a separate uniform-integrability or convergence argument.

6 Girsanov Moves Drift into the Measure

6.1 The theorem and its sign convention

Theorem 6.1 (Girsanov, one-dimensional version)**.** Let

$$Z_t = \exp\left(\int_0^t \phi_s \, \mathrm{d}W_s - \frac{1}{2}\int_0^t \phi_s^2 \, \mathrm{d}s\right)$$

be a true $\mathbb{P}$-martingale, and define

$$\left.\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}\right|_{\mathcal{F}_T} = Z_T.$$

Then

$$\boxed{W_t^{\mathbb{Q}} = W_t - \int_0^t \phi_s \, \mathrm{d}s}$$

is Brownian motion under $\mathbb{Q}$.

Always solve this identity for the old differential before substituting:

$$\boxed{\mathrm{d}W_t = \mathrm{d}W_t^{\mathbb{Q}} + \phi_t \, \mathrm{d}t.}$$

Hence, if under $\mathbb{P}$

$$\mathrm{d}X_t = b_t \, \mathrm{d}t + a_t \, \mathrm{d}W_t,$$

then under $\mathbb{Q}$

$$\boxed{\mathrm{d}X_t = (b_t + a_t\phi_t) \, \mathrm{d}t + a_t \, \mathrm{d}W_t^{\mathbb{Q}}.}$$

Only the drift changes. The diffusion coefficient, and therefore the local quadratic variation, is unchanged:

$$\mathrm{d}[X]_t = a_t^2 \, \mathrm{d}t$$

under both measures.

A sign table worth memorizing

density exponent	new Brownian motion	new drift of X
$+\int \phi \, \mathrm{d}W - \frac{1}{2}\int \phi^2 \, \mathrm{d}t$	$W^{\mathbb{Q}} = W - \int \phi \, \mathrm{d}t$	$b + a\phi$
$-\int \lambda \, \mathrm{d}W - \frac{1}{2}\int \lambda^2 \, \mathrm{d}t$	$W^{\mathbb{Q}} = W + \int \lambda \, \mathrm{d}t$	$b - a\lambda$

The second row is the form most often used in asset pricing.

Example — remove a deterministic drift

Suppose

$$\mathrm{d}X_t = \alpha_t \, \mathrm{d}t + \beta_t \, \mathrm{d}W_t, \qquad \beta_t \neq 0,$$

with deterministic coefficients. To make X driftless, choose

$$\phi_t = -\frac{\alpha_t}{\beta_t}.$$

Then, provided the exponential is a true martingale,

$$\mathrm{d}X_t = \beta_t \, \mathrm{d}W_t^{\mathbb{Q}}.$$

This is first an algebraic candidate. Its validity still depends on the density.

6.2 Write the density in each drift-change example

For $\mathrm{d}X_t = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}W_t$, $\sigma > 0$, target drift ν under $\mathbb{Q}$ by taking

$$\lambda = \frac{\mu - \nu}{\sigma}, \qquad Z_t = e^{-\lambda W_t - \lambda^2 t/2}, \qquad \left.\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}\right|_{\mathcal{F}_T} = Z_T.$$

Then $W_t^{\mathbb{Q}} = W_t + \lambda t$ and $\mathrm{d}X_t = \nu\,\mathrm{d}t + \sigma\,\mathrm{d}W_t^{\mathbb{Q}}$.
For the adapted example $\mathrm{d}X_t = \sin(W_t)\,\mathrm{d}t + \mathrm{d}W_t$, $X_0 = 0$, set

$$Z_t = \exp\left(-\int_0^t \sin(W_s)\,\mathrm{d}W_s - \frac{1}{2}\int_0^t \sin^2(W_s)\,\mathrm{d}s\right), \qquad \left.\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}\right|_{\mathcal{F}_T} = Z_T.$$

The integrand is bounded, so Novikov applies. The process $W_t^{\mathbb{Q}} = W_t + \int_0^t \sin(W_s)\,\mathrm{d}s$ is Brownian under $\mathbb{Q}$, and $\mathrm{d}X_t = \mathrm{d}W_t^{\mathbb{Q}}$.
For a geometric diffusion $\mathrm{d}S_t/S_t = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}W_t$, use the same $\lambda = (\mu - \nu)/\sigma$ and the full density

$$Z_t = e^{-\lambda W_t - \lambda^2 t/2}, \qquad \left.\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}\right|_{\mathcal{F}_T} = Z_T.$$

It yields $\mathrm{d}S_t/S_t = \nu\,\mathrm{d}t + \sigma\,\mathrm{d}W_t^{\mathbb{Q}}$. For the money-market pricing measure, the target is $\nu = r$.

6.3 Conditional expectations and path-dependent claims

For any $Y \in L^1(\mathbb{Q})$,

$$\mathbb{E}^{\mathbb{Q}}[Y \mid \mathcal{F}_t] = \frac{1}{Z_t}\mathbb{E}^{\mathbb{P}}[Z_T Y \mid \mathcal{F}_t].$$

This identity remains valid when Y depends on the whole path. For example,

$$Y = \mathbf{1}_{\{\max_{0\le u\le T} W_u > K\}}$$

does not become a function of W_T merely because the measure changes. Girsanov changes the weight of each path, not the information contained in the payoff.
In the geometric example with constant ν, σ, the two routes give the same conditional mean:

$$\boxed{\mathbb{E}^{\mathbb{Q}}[S_T \mid \mathcal{F}_t] = S_t e^{\nu(T-t)} = \frac{1}{Z_t}\mathbb{E}^{\mathbb{P}}[Z_T S_T \mid \mathcal{F}_t].}$$

Three checkpoints for every Girsanov problem

1. **Density:** write the linear stochastic integral and the negative half-energy compensator with the intended sign.
2. **Validity:** prove that the exponential is a true martingale, or clearly state the assumption that guarantees it.
3. **Substitution:** identify the new Brownian motion, solve for the old $\mathrm{d}W$, and only then rewrite the SDE.

7 The Risk-Neutral Measure in Black–Scholes

7.1 Constructing the pricing density

Under $\mathbb{P}$, assume

$$\frac{\mathrm{d}S_t}{S_t} = \mu_t\,\mathrm{d}t + \sigma_t\,\mathrm{d}W_t, \qquad \mathrm{d}B_t = r_t B_t\,\mathrm{d}t, \qquad \sigma_t \neq 0.$$

Define

$$\lambda_t = \frac{\mu_t - r_t}{\sigma_t}.$$

The pricing-density candidate is

$$\boxed{Z_t = \exp\Big(-\int_0^t \lambda_s \, \mathrm{d}W_s - \frac{1}{2}\int_0^t \lambda_s^2 \, \mathrm{d}s\Big).}$$

Assume it is a true martingale and set $\mathrm{d}\mathbb{Q}/\,\mathrm{d}\mathbb{P}|_{\mathcal{F}_T} = Z_T$. Girsanov gives

$$\boxed{W_t^{\mathbb{Q}} = W_t + \int_0^t \lambda_s \, \mathrm{d}s, \qquad \mathrm{d}W_t = \mathrm{d}W_t^{\mathbb{Q}} - \lambda_t \, \mathrm{d}t.}$$

Substituting into the stock SDE,

$$\begin{aligned}\frac{\mathrm{d}S_t}{S_t} &= \mu_t \, \mathrm{d}t + \sigma_t(\,\mathrm{d}W_t^{\mathbb{Q}} - \lambda_t \, \mathrm{d}t) \\ &= (\mu_t - \sigma_t\lambda_t)\,\mathrm{d}t + \sigma_t \, \mathrm{d}W_t^{\mathbb{Q}} \\ &= \boxed{r_t \, \mathrm{d}t + \sigma_t \, \mathrm{d}W_t^{\mathbb{Q}}.}\end{aligned}$$

The statistical drift μ_t has been replaced by the short rate, while the volatility remains unchanged.

Let

$$D_t = B_t^{-1} = \exp\Big(-\int_0^t r_s \, \mathrm{d}s\Big).$$

Under $\mathbb{Q}$,

$$\boxed{\mathrm{d}(D_tS_t) = \sigma_t D_t S_t \, \mathrm{d}W_t^{\mathbb{Q}}.}$$

Thus (DS) is a local martingale and, under suitable integrability, a true martingale.

7.2 A call formula from two Gaussian tails

For constant r, σ and $\tau = T - t$, the risk-neutral SDE gives

$$S_T = S_t \exp\Big((r - \tfrac{1}{2}\sigma^2)\tau + \sigma(W_T^{\mathbb{Q}} - W_t^{\mathbb{Q}})\Big).$$

Write

$$d_- = \frac{\log(S_t/K) + (r - \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}, \qquad d_+ = d_- + \sigma\sqrt{\tau}.$$

The cash-or-nothing tail and the stock-weighted tail are

$$\mathbb{Q}(S_T > K \mid \mathcal{F}_t) = \Phi(d_-), \qquad e^{-r\tau}\mathbb{E}^{\mathbb{Q}}[S_T \mathbf{1}_{\{S_T>K\}} \mid \mathcal{F}_t] = S_t\Phi(d_+).$$

Since $(S_T - K)^+ = S_T\mathbf{1}_{\{S_T>K\}} - K\mathbf{1}_{\{S_T>K\}}$,

$$\boxed{C_t = S_t\Phi(d_+) - Ke^{-r\tau}\Phi(d_-), \qquad \Delta_t = \Phi(d_+).}$$

The pricing identity and the replication argument behind it are developed in Lecture 7.

Exercise — catch the wrong pricing sign

For constant $\lambda = (\mu - r)/\sigma$, suppose one mistakenly uses

$$\widetilde{Z}_T = e^{+\lambda W_T - \lambda^2 T/2}.$$

Find the stock drift under the corresponding measure.

Short solution

Now $\widetilde{W}_t = W_t - \lambda t$ is Brownian, so $\mathrm{d}W_t = \mathrm{d}\widetilde{W}_t + \lambda \, \mathrm{d}t$. Therefore

$$\frac{\mathrm{d}S_t}{S_t} = (\mu + \sigma\lambda)\,\mathrm{d}t + \sigma \, \mathrm{d}\widetilde{W}_t = (2\mu - r)\,\mathrm{d}t + \sigma \, \mathrm{d}\widetilde{W}_t.$$

The discounted drift is $2(\mu - r)$, so the sign is wrong.

7.3 What "risk-neutral" does and does not mean

Definition 7.1 (Equivalent martingale measure)**.** Relative to the positive numeraire B, a probability measure $\mathbb{Q}$ is an equivalent martingale measure if

1. $\mathbb{Q} \sim \mathbb{P}$ on $\mathcal{F}_T$, and
2. every discounted traded price S^i/B is a $\mathbb{Q}$-martingale under the required admissibility and integrability conditions.

The term "risk-neutral" means that traded assets have drift r_t when expressed in currency units, equivalently zero drift after discounting. It does not mean:

- that real investors are indifferent to risk;
- that $\mathbb{Q}$ is the empirical distribution of future returns;
- that all payoffs have expected return r under $\mathbb{P}$;
- that changing measure removes randomness.

Why the physical drift disappears from option prices

Replication and Girsanov express the same no-arbitrage restriction in two languages. Delta hedging cancels the stock's Brownian exposure and therefore its associated physical drift. Girsanov reweights paths so that discounted traded assets are martingales. Both routes produce the same pricing PDE and the same value.

8 Guided Practice and Common Traps

8.1 A complete change-of-measure drill

Exercise — a constant Brownian shift

Let W be Brownian motion under $\mathbb{P}$, fix $a \in \mathbb{R}$, and define

$$Z_T = e^{-aW_T - a^2T/2}, \qquad \frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}} = Z_T.$$

1. Verify $\mathbb{E}^{\mathbb{P}}[Z_T] = 1$.
2. Identify the $\mathbb{Q}$-Brownian motion.
3. Find the law of W_T under $\mathbb{Q}$.
4. Rewrite $\mathrm{d}X_t = b\,\mathrm{d}t + c\,\mathrm{d}W_t$ under $\mathbb{Q}$.

Short solution

The Gaussian mgf gives the normalization. Here $\phi = -a$, so

$$W_t^{\mathbb{Q}} = W_t + at, \qquad W_T = W_T^{\mathbb{Q}} - aT.$$

Hence

$$W_T \sim N(-aT, T) \quad \text{under } \mathbb{Q},$$

and $\mathrm{d}W_t = \mathrm{d}W_t^{\mathbb{Q}} - a\,\mathrm{d}t$ yields

$$\boxed{\mathrm{d}X_t = (b - ac)\,\mathrm{d}t + c\,\mathrm{d}W_t^{\mathbb{Q}}.}$$

8.2 Diagnose candidate densities

Exercise — which conclusions are justified?

Consider

candidate	available information
$e^{\theta W_T-\theta^2T/2}$	$\theta\in\mathbb{R}$
$e^{\int_0^T \phi_s\,\mathrm{d}W_s-\frac{1}{2}\int_0^T \phi_s^2\,\mathrm{d}s}$	$\lvert\phi_s\rvert\le C$
$e^{\int_0^T \psi_s\,\mathrm{d}W_s-\frac{1}{2}\int_0^T \psi_s^2\,\mathrm{d}s}$	$\int_0^T \psi_s^2\,\mathrm{d}s<\infty$ a.s.

For each row, decide whether the stated information is enough to define a probability measure.

Short solution

The first density has expectation one by the Gaussian mgf. The second satisfies Novikov because the integrand is bounded. The third is only known to be a positive local martingale; finite energy almost surely does not imply the exponential moment required by Novikov or any equivalent criterion.

8.3 Error diagnosis

Exercise — correct six statements

Correct the following claims:

1. A change of measure changes the realized payoff $H(\omega)$.
2. $Z_T\ge 0$ is enough to define a probability measure.
3. The conditional change-of-measure formula has denominator Z_T.
4. Zero drift of a stochastic exponential proves it is a true martingale.
5. Girsanov changes both drift and volatility.
6. Under a risk-neutral measure, every undiscounted traded price is a martingale.

Short solution

1. The payoff and paths stay fixed; only scenario weights change.
2. One also needs $\mathbb{E}^{\mathbb{P}}[Z_T]=1$; strict positivity gives equivalence.
3. It is $Z_t=\mathbb{E}^{\mathbb{P}}[Z_T\mid\mathcal{F}_t]$.
4. Zero drift gives a local martingale first; validity requires a true-martingale argument.
5. The drift changes, while diffusion and quadratic variation remain.
6. Prices discounted by the chosen numeraire are martingales.

8.4 The Black–Scholes sign drill

Exercise — five lines from the historical model

Assume constant r,μ,σ, with $\sigma>0$, and $\mathrm{d}S_t/S_t=\mu\,\mathrm{d}t+\sigma\,\mathrm{d}W_t^{\mathbb{P}}$. Write, in order, the market price of risk, density process, $\mathbb{Q}$-Brownian motion, $\mathbb{Q}$-stock dynamics, and discounted-stock dynamics.

Short solution

$$\lambda=\frac{\mu-r}{\sigma},\qquad Z_t=e^{-\lambda W_t^{\mathbb{P}}-\lambda^2t/2},\qquad \left.\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}\right|_{\mathcal{F}_T}=Z_T,$$

$$W_t^{\mathbb{Q}}=W_t^{\mathbb{P}}+\lambda t,\qquad \frac{\mathrm{d}S_t}{S_t}=r\,\mathrm{d}t+\sigma\,\mathrm{d}W_t^{\mathbb{Q}},\qquad \boxed{\mathrm{d}(e^{-rt}S_t)=\sigma e^{-rt}S_t\,\mathrm{d}W_t^{\mathbb{Q}}.}$$

The last expression is a stochastic integral and hence a local martingale; calling it a true martingale needs an integrability argument.

8.5 End-of-lecture checklist

Nine checks for a change-of-measure calculation

1. State the original measure, filtration, and horizon.
2. Identify the drift that must change and the diffusion that cannot.
3. Write the candidate stochastic exponential with its compensator.
4. Check nonnegativity, unit expectation, and equivalence.
5. Use Novikov, boundedness, localization, or another valid criterion.
6. State the new Brownian motion with the correct sign.
7. Solve for the old differential before substituting into the SDE.
8. When conditioning, use Z_t, not the terminal density alone.
9. For pricing, name the numeraire and discount every traded asset.

Final synthesis

- A density reweights unchanged scenarios, and Bayes' rule requires conditional renormalization.
- Gaussian tilting changes a mean without changing variance; Girsanov is its pathwise Brownian analogue.
- The half-quadratic-variation term makes a stochastic exponential a local martingale, not automatically a true martingale.
- Girsanov moves drift into the probability measure while preserving diffusion and quadratic variation.
- The negative market-price-of-risk density replaces μ by r and makes the discounted stock a local martingale. Lecture 7 develops the integrability, replication, and completeness steps needed for general valuation.